

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)**

II B.Tech II Semester Regular Examinations, April - 2025

PROBABILITY AND STATISTICS

(Common to CSE, IT, CSE(AI&ML) and CSE(AI))

Time: 3 hours

Max. Marks: 70

Note:

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.	Questions		BTL	CO	Marks																		
1.	a)	What are the major limitations of Descriptive statistics?	K1	CO1	2M																		
	b)	Define interval and ratio variables.	K2	CO1	2M																		
	c)	Write the formulae for rank correlation for both tied and untied ranks.	K2	CO2	2M																		
	d)	Define multiple linear regression.	K1	CO2	2M																		
	e)	Define probability and Conditional probability.	K2	CO3	2M																		
	f)	If z is a standard normal variate, then find the area to the left of $z = -1.78$	K2	CO3	2M																		
	g)	Define sampling distribution of variance.	K2	CO4	2M																		
	h)	Write the formula to find the confidence interval for true mean in case of large sample.	K2	CO4	2M																		
	i)	Find the values of (i) $F_{0.05}$ with $\nu_1 = 7$ and $\nu_2 = 15$ (ii) $F_{0.01}$ with $\nu_1 = 28$ and $\nu_2 = 12$.	K2	CO5	2M																		
	j)	Briefly explain Chi-squared test.	K2	CO5	2M																		
UNIT – I																							
2.	a)	Find the median of the following data : <table><tr><td>Class interval</td><td>5-10</td><td>10-15</td><td>15-20</td><td>20-25</td><td>25-30</td><td>30-35</td><td>35-40</td><td>40-45</td></tr><tr><td>Frequency</td><td>5</td><td>6</td><td>15</td><td>10</td><td>5</td><td>4</td><td>2</td><td>2</td></tr></table>	Class interval	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45	Frequency	5	6	15	10	5	4	2	2	K3	CO1	5M
	Class interval	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45														
	Frequency	5	6	15	10	5	4	2	2														
b)	The frequency distribution of weights of a group of 60 salesmen of a company is given below: <table><tr><td>Weights (in Kgs.)</td><td>50-54</td><td>55-59</td><td>60-64</td><td>65-69</td><td>70-74</td><td>75-79</td><td>80-84</td></tr><tr><td>No. of Salesmen</td><td>6</td><td>10</td><td>24</td><td>36</td><td>28</td><td>12</td><td>4</td></tr></table> Draw Histogram for the above distribution and hence find the mode.	Weights (in Kgs.)	50-54	55-59	60-64	65-69	70-74	75-79	80-84	No. of Salesmen	6	10	24	36	28	12	4	K3	CO1	5M			
Weights (in Kgs.)	50-54	55-59	60-64	65-69	70-74	75-79	80-84																
No. of Salesmen	6	10	24	36	28	12	4																
OR																							
3.	a)	What are the main methods of collecting primary data? Explain them.	K2	CO1	5M																		
	b)	Calculate the variance and standard deviation of the following data: <table><tr><td>Class interval</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td></tr><tr><td>Frequency</td><td>8</td><td>13</td><td>16</td><td>8</td><td>5</td></tr></table>	Class interval	0-10	10-20	20-30	30-40	40-50	Frequency	8	13	16	8	5	K3	CO1	5M						
	Class interval	0-10	10-20	20-30	30-40	40-50																	
Frequency	8	13	16	8	5																		
UNIT – II																							
4.	a)	Find the correlation coefficient for the following data <table><tr><td>x</td><td>12</td><td>9</td><td>8</td><td>10</td><td>11</td><td>13</td><td>7</td></tr><tr><td>y</td><td>14</td><td>8</td><td>6</td><td>9</td><td>11</td><td>12</td><td>3</td></tr></table>	x	12	9	8	10	11	13	7	y	14	8	6	9	11	12	3	K3	CO2	5M		
	x	12	9	8	10	11	13	7															
	y	14	8	6	9	11	12	3															
b)	Fit the second degree polynomial $y = a + bx + cx^2$ to the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1</td><td>18</td><td>13</td><td>25</td><td>63</td></tr></table>	x	0	1	2	3	4	y	1	18	13	25	63	K3	CO2	5M							
x	0	1	2	3	4																		
y	1	18	13	25	63																		
OR																							
5.	a)	Find the line of regression of y on x from the following data: <table><tr><td>x</td><td>10</td><td>20</td><td>25</td><td>13</td><td>16</td><td>25</td></tr><tr><td>y</td><td>25</td><td>18</td><td>24</td><td>25</td><td>12</td><td>17</td></tr></table>	x	10	20	25	13	16	25	y	25	18	24	25	12	17	K3	CO2	5M				
	x	10	20	25	13	16	25																
	y	25	18	24	25	12	17																

	b)	Fit a curve of the form $y = ab^x$ to the following data using method of least squares.		K3	CO2	5M																
		<table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>5.0</td><td>6.3</td><td>7.2</td><td>7.9</td><td>8.5</td></tr></table>	x	1	2	3	4	5	y	5.0	6.3	7.2	7.9	8.5								
x	1	2	3	4	5																	
y	5.0	6.3	7.2	7.9	8.5																	
UNIT – III																						
6.	a)	A random variable X has the following distribution:		K3	CO3	5M																
		<table><tr><td>X=x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>P(X=x)</td><td>k</td><td>3k</td><td>5k</td><td>7k</td><td>9k</td><td>11k</td><td>13k</td></tr></table>	X=x	0	1	2	3	4	5	6	P(X=x)	k	3k	5k	7k	9k	11k	13k				
X=x	0	1	2	3	4	5	6															
P(X=x)	k	3k	5k	7k	9k	11k	13k															
		Then find (i) k and (ii) Mean.																				
	b)	If a Poisson distribution is such that $\frac{3}{2}P(X=1) = P(X=3)$ then find $P(2 \leq X \leq 5)$.		K3	CO3	5M																
OR																						
7.	a)	The probabilities of A, B and C to become Vice-Chancellor of a university are 0.5, 0.3 and 0.2 respectively. The probabilities that the bonus scheme will be introduced if they become Vice-chancellor are 0.2, 0.3 and 0.4 respectively. Determine the probability that C become Vice-chancellor if the bonus scheme is introduced.		K3	CO3	5M																
	b)	Suppose the weights of 800 students are normally distributed with mean 140 pounds and standard deviation 10 pounds. Find the number of students whose weights are between 138 and 148 pounds.		K3	CO3	5M																
UNIT – IV																						
8.	a)	Construct 90% confidence interval for population mean, where sample mean $\bar{x} = 53$, Standard deviation $\sigma = 7.1$ and sample size $n = 90$.		K3	CO4	5M																
	b)	A sample of size 100 is taken from an infinite population having the mean 75 and variance 256. What is the probability that the sample mean will be between 67 and 83.		K3	CO4	5M																
OR																						
9.		Let the elements of the population be 1, 3, 5, 7, 9. Samples of size 2 is taken without replacement. Then (i) Find the Population mean (ii) Find the Population standard deviation (iii) List all samples of size 2 without replacement (iv) Find the mean of sampling distribution of means (v) Find the standard deviation of sampling distribution of means.		K3	CO4	10M																
UNIT – V																						
10.	a)	A trucking firm suspects the claim that the average life time of certain tires is at least 28000 miles. To check the claim, the firm puts 40 of these tires on its trucks and gets a mean life time of 27463 miles with a standard deviation of 1348 miles. Test the claim at 0.01 level of significance.		K3	CO5	5M																
	b)	An oil company claims that less than 20% of all car owners have not tried its gasoline. Test this claim at 0.05 level of significance, if a random check reveals that 22 of 200 car owners have not tried the oil company's gasoline.		K3	CO5	5M																
OR																						
11.	a)	A random sample of 8 envelopes is taken from the letter box of a post office and their weights in grams are found to be 12.1, 11.9, 12.4, 12.3, 11.5, 11.6, 12.1 and 12.4 grams. Does this sample indicate at 0.01 level of significance that the average weight of envelopes received at their post office is 12.35 grams.		K3	CO5	5M																
	b)	A researcher wants to know the intelligence of students in a school. He selected two groups of students. In the first group, there are 150 students having mean IQ of 75 with standard deviation of 15 and in the second group, there are 250 students having mean IQ of 70 with standard deviation of 20. Is there a significant difference between the means of two groups?		K3	CO5	5M																